

BASIC MACHINE LEARNING CLASSIFICATION

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BASIC MACHINE LEARNING CLASSIFICATION



K-NEAREST NEIGHBOR (KNN)

- Is a supervised machine learning algorithm
- It works by **finding “k” closest data** point (neighbors) to given input and makes predictions based on the majority class (for classification) or average value (for regression).
- Lazy learner → it doesn't learn from the training set immediately instead it stores the dataset and at the time of classification it performs an action on the dataset
- Distance metrics used in kNN Algorithm

- Euclidean Distance

$$\text{Euclidean Distance} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

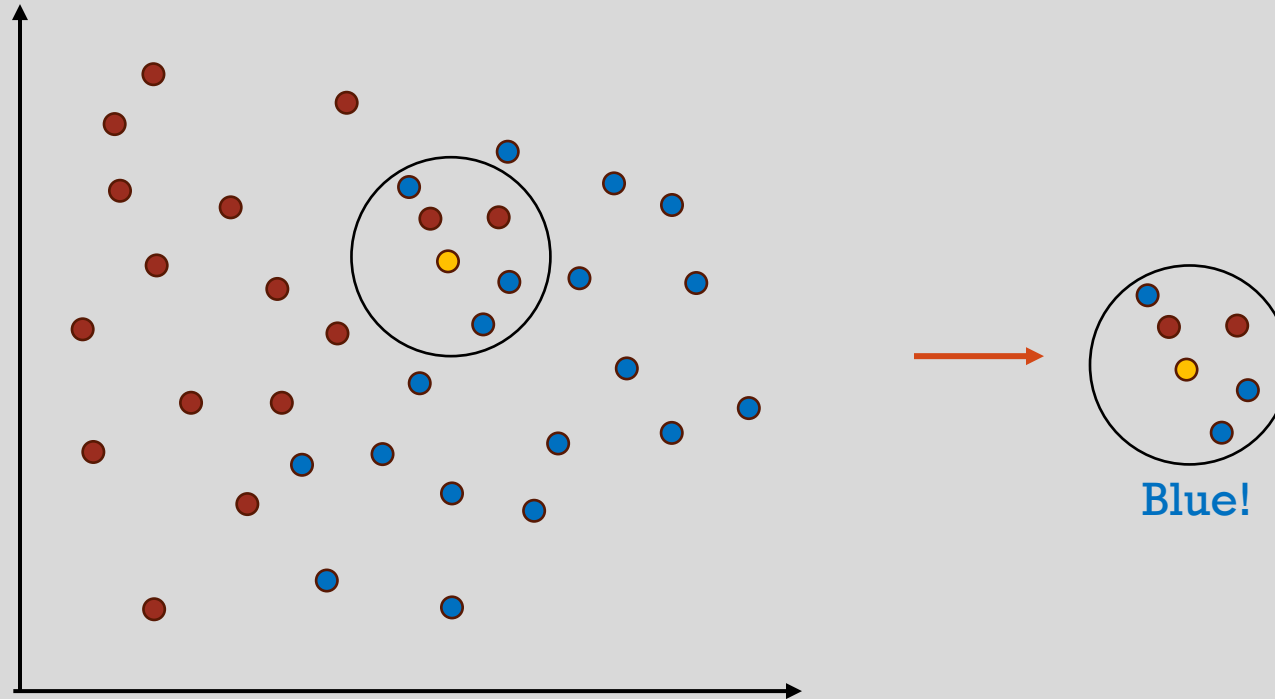
- Manhattan Distance

$$\text{Manhattan Distance} = |x_1 - x_2| + |y_1 - y_2|$$

- Minkowski Distance

$$\text{Minkowski Distance} = \sqrt[p]{(x_1 - x_2)^p + (y_1 - y_2)^p}$$

K-NEAREST NEIGHBOR (KNN)



Pros	Cons
• Easy to understand • Lazy learner	• Slow if the data is big • Sensitive to scale and noise

NAÏVE BAYES

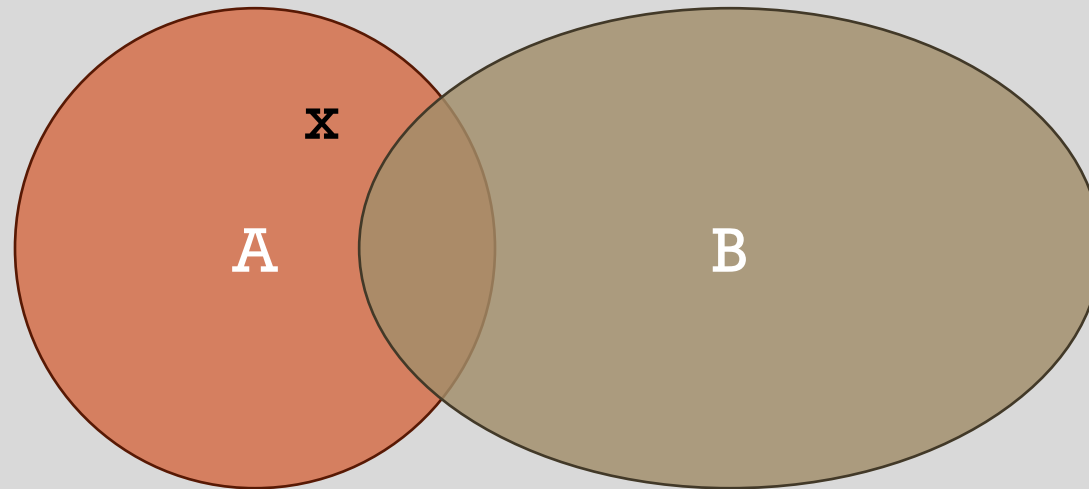
- Naïve Bayes is a probabilistic classifier based on Bayes' Theorem.
- It **assumes that all features are independent** given the class → “naïve”

$$P(Class | Data) = \frac{P(Data | Class) \cdot P(Class)}{P(Data)}$$

- $P(Class | Data)$: Probability of the class given the image data
- $P(Data | Class)$: Probability of seeing this data if we already know the class
- $P(Class)$: How common this class is
- $P(Data)$: Overall probability of the data

NAÏVE BAYES

- Classes A and B as two sets:
 - $P(A | x)$: the probability of A is observed when seeing an x
 - $P(B | x)$: the probability of B is observed when seeing an x



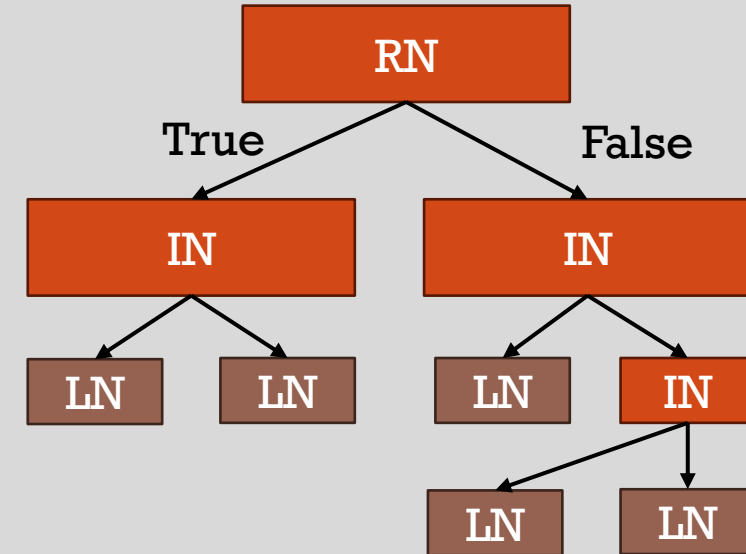
NAÏVE BAYES

- Type of Naïve Bayes Model:
 - Gaussian Naïve Bayes
 - Multinomial Naïve Bayes
 - Bernoulli Naïve Bayes

Pros	Cons
• Fast and efficient • Extendable to multi-class problems • Requires less training examples • Works well for categorical data	• Features are assumed to be independent to each other (not true in real-world applications) • Zero-frequency problem

DECISION TREE

- A decision tree is a model that **splits data into branches** based on feature values, like a flowchart.
- It ask **yes/no questions** and ends in a decision (class label)
- How it works:
 - Selects the best feature to split the data at each node
 - Continues splitting until it can classify the data or reaches a limit

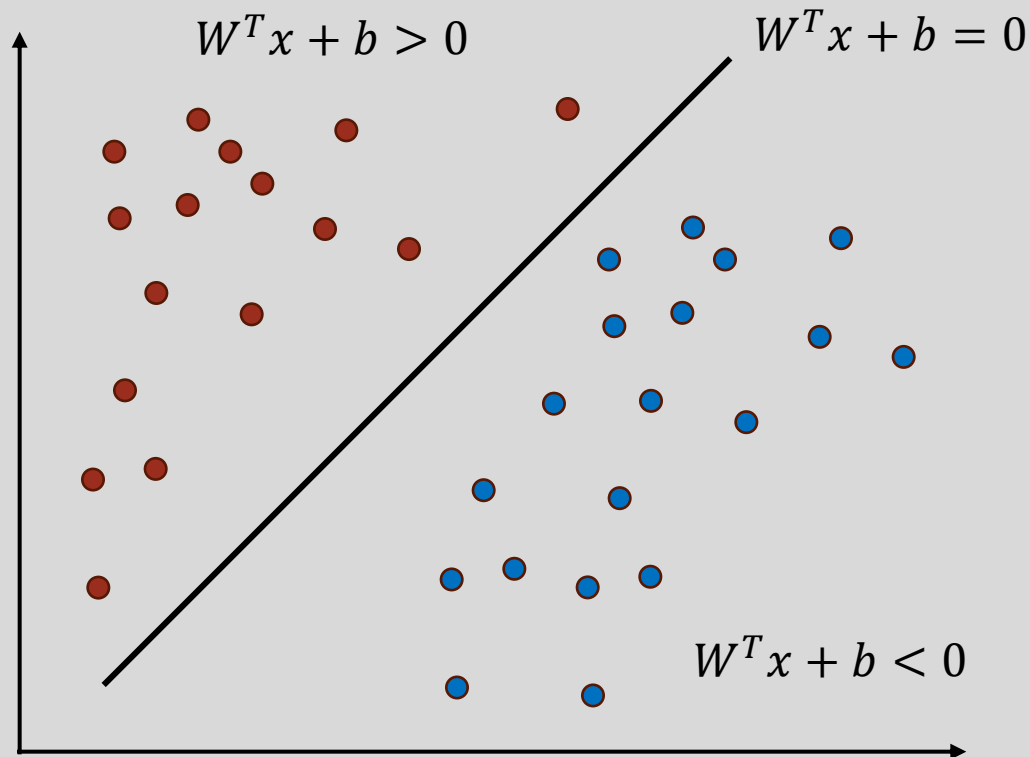


Pros	Cons
• Easy to visualize and explain. • No need to normalize data. • Works well with categorical and numerical data.	• Can overfit if not pruned (makes a tree too deep). • Not great for high-dimensional, noisy image data.

RN: Root Node
IN: Internal Node/Branches
LN: Leaf Node/Leaves

SUPPORT VECTOR MACHINES (SVM)

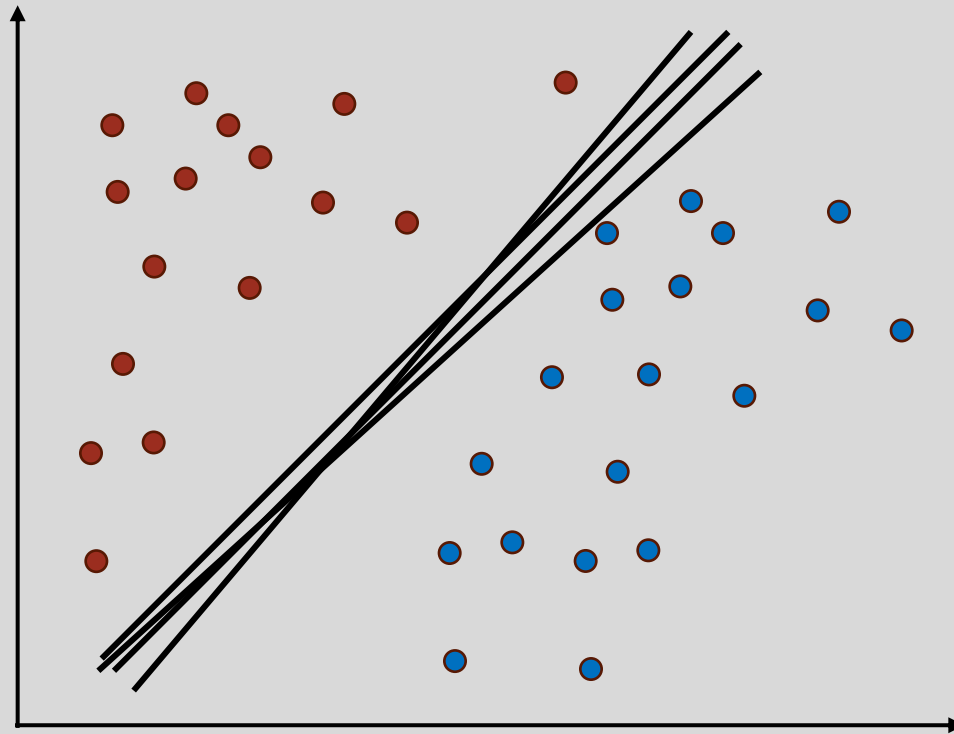
- Binary classification can be viewed as the task of separating classes in feature space:



$$f(x) = \text{sign}(W^T x + b)$$

SUPPORT VECTOR MACHINES (SVM)

- Binary classification can be viewed as the task of separating classes in feature space:



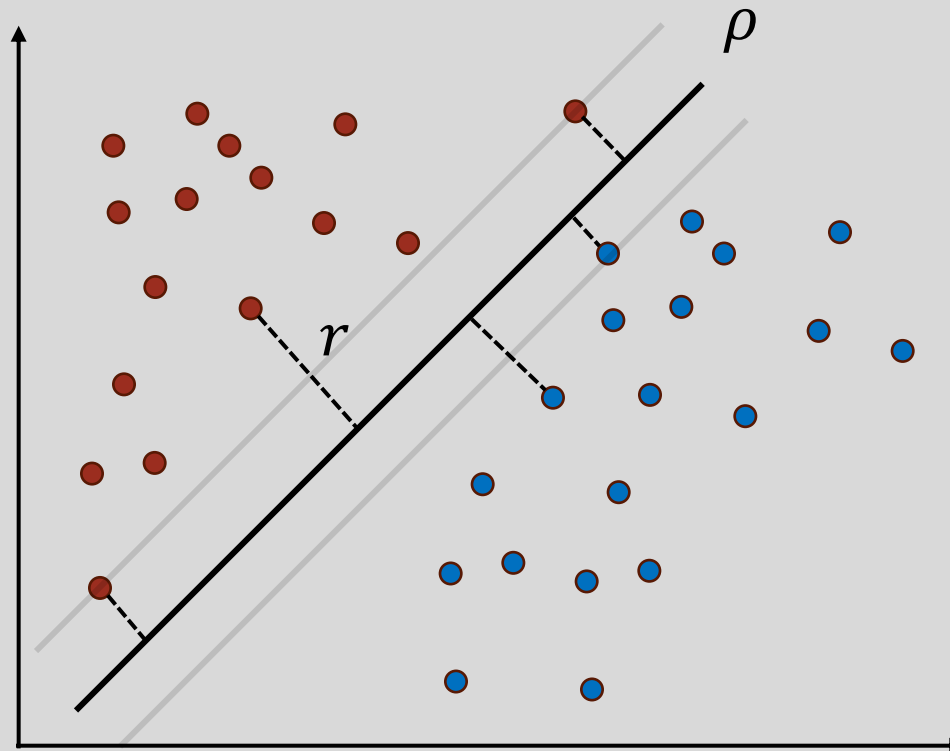
Which one is the **best**?

SUPPORT VECTOR MACHINES (SVM) - MARGIN

- Distance from sample x_i to the separator is

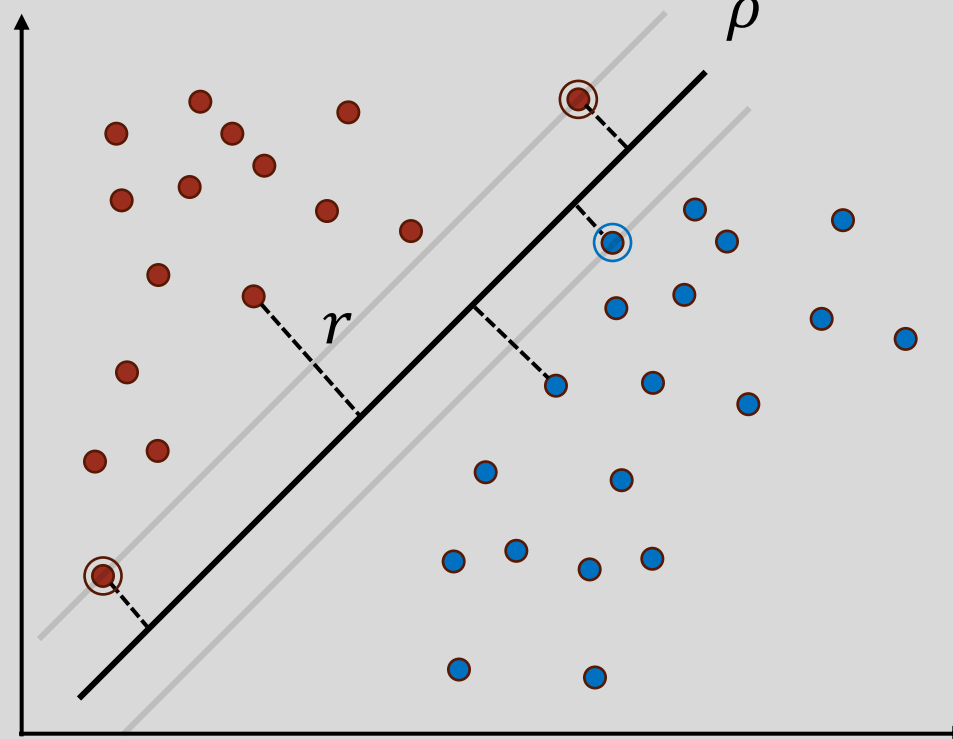
$$r = \frac{w^T x_i + b}{\|b\|}$$

- Examples closest to the hyperplane are support vectors
- Margin** is the shortest distance between support vectors and the hyperplane



SUPPORT VECTOR MACHINES (SVM) - MARGIN

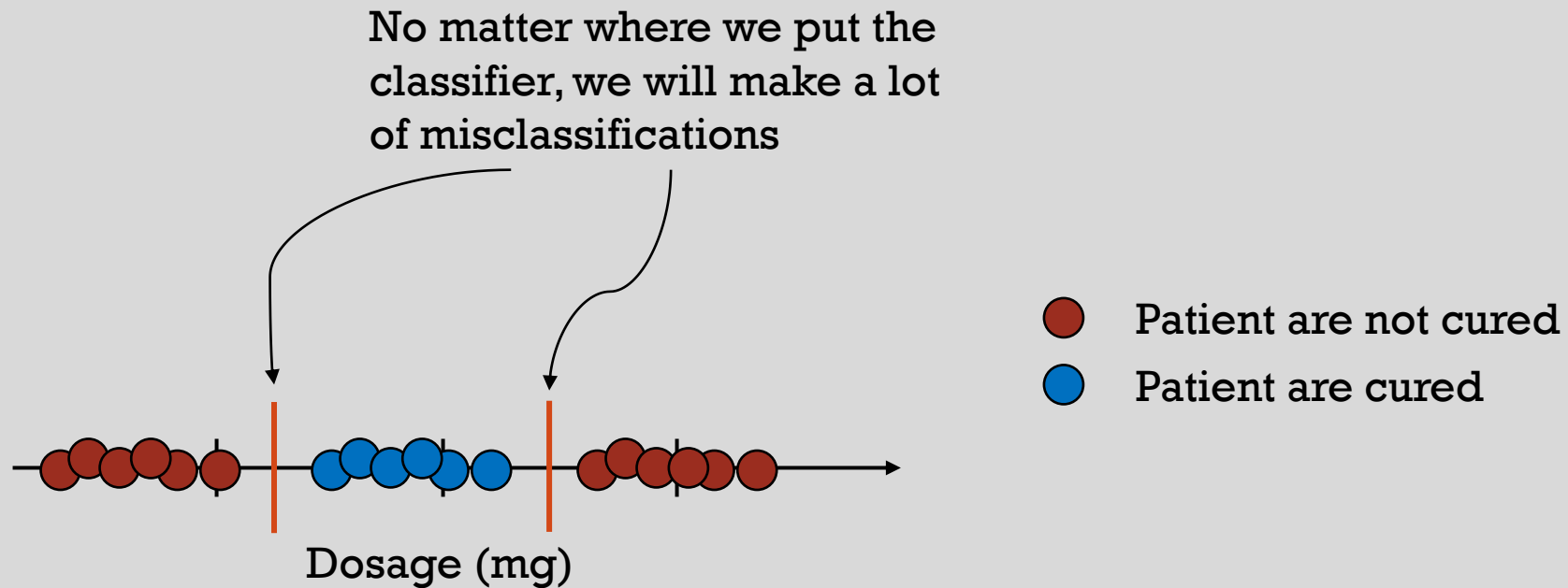
- Maximizing the margin is good according to intuition and PAC theory (Probably Approximately Correct) → **maximal margin classifier**
- Maximal margin classifier are **super sensitive to outliers** in the training data
- **Soft margin** → the distance of the support vectors and the threshold that is not so sensitive to outliers, so it will **allow some misclassifications**



Use cross validation to determine how many misclassification to allow inside of the soft margin to get the best classification

SUPPORT VECTOR MACHINES (SVM)

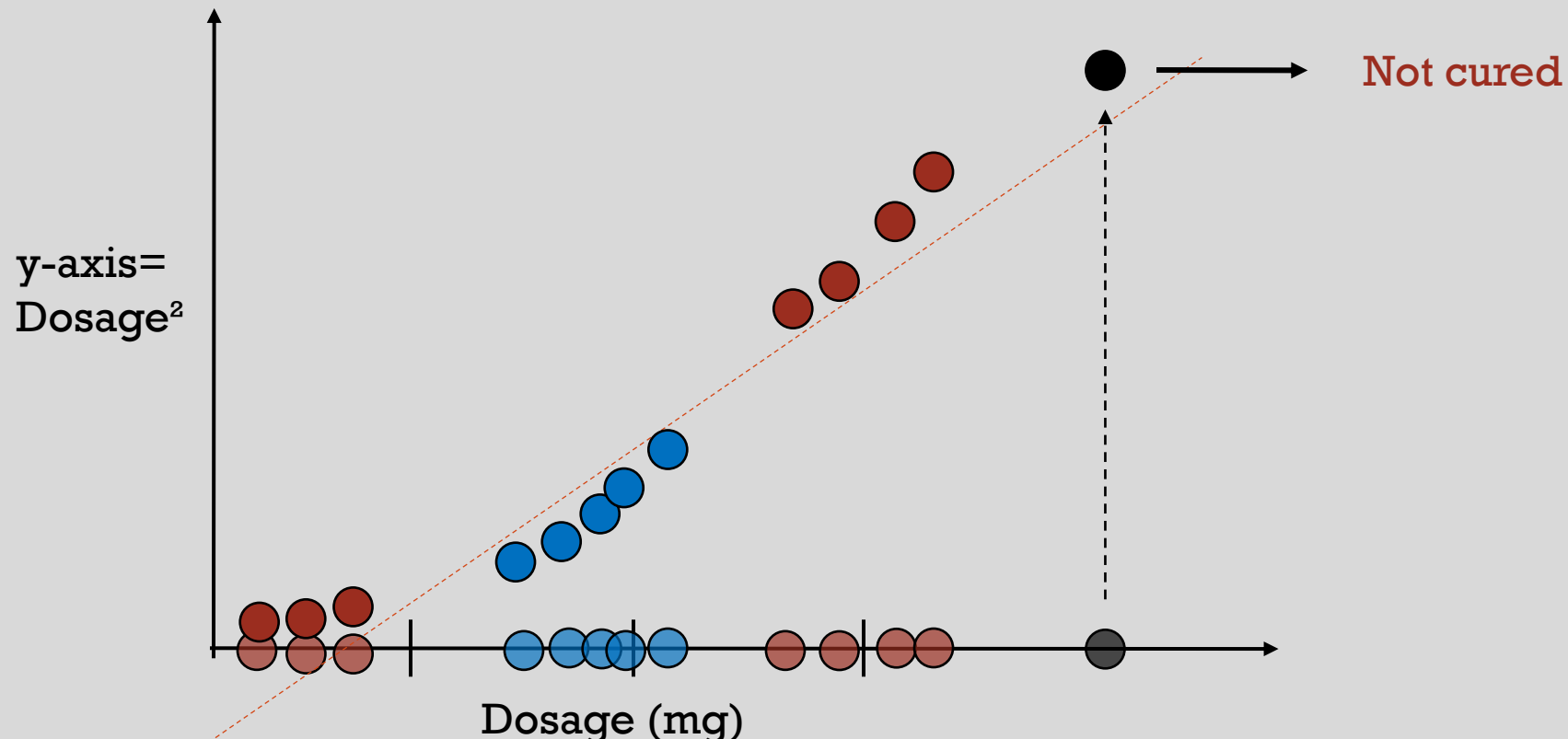
- When we use soft margin to determine the location of the threshold/hyperplane → we are using **soft margin classifier/ support vector classifier** to classify observations



Support Vector
Machines (SVM)

SUPPORT VECTOR MACHINES (SVM)

- Main idea of SVM:
 1. Start with data in a relatively low dimension
 2. Move the data into higher dimension
 3. Find the support vector classifier that separates the higher data into two groups



SUPPORT VECTOR MACHINES (SVM)

- SVM is a supervised learning model that tries to find the best boundary (hyperplane) that separates classes with the widest possible margin
- In order to make the mathematics possible, SVM use **kernel functions** to systematically find support vector classifiers in higher dimensions
- The **polynomial kernel** systematically increases dimensions by setting d , the degree of the polynomial → relationship between each pair of observations are used to **find support vector classifier**

Pros	Cons
• High performance on small-to-medium datasets. • Effective in high-dimensional spaces (like images). • Works well even when classes are not linearly separable (with kernel trick).	• Not well suited for very large datasets (slow training). • Choice of kernel and parameters affects performance.

THANK YOU